

Grounded DI LLC - Rocket AnalyzerNew Glenn LC-36 Static-Fire Anomaly

Geometry-Origin Audit + Improved Public Artifact*Prepared May 29, 2026 | Public-source visual diagnostic | Not a root-cause finding***1. Executive Audit Result**

Defensible revised finding	Earliest visible luminous development appears low in the aft region, slightly to camera-right of the vehicle centerline, adjacent to the vehicle / pad interface band.
Revised geometry class	B/C interface band: on or immediately outside the lower-right projected vehicle boundary from the camera perspective.
Root cause	UNRESOLVED. Public imagery cannot distinguish vehicle aft plumbing, engine-bay initiation, umbilical/QD hardware, pad-side propellant services, or secondary trench/ground combustion with sufficient confidence.
Confidence posture	Moderate confidence for camera-plane lower-aft interface localization; low confidence for assigning vehicle-side vs ground-side origin; zero confidence for root-cause assignment without telemetry and investigation data.

2. Public Fact Baseline

- Event: Blue Origin New Glenn anomaly during a static-fire / hotfire test at Launch Complex 36, Cape Canaveral Space Force Station, on May 28, 2026, around 9:00 p.m. ET.
- Personnel: public reporting and Blue Origin statements reported all personnel accounted for; no injuries or fatalities were reported.
- Payload: public reporting indicates the 48 Amazon Leo satellites were not aboard the vehicle during the event.
- Status: Blue Origin investigation / damage assessment underway; official root cause not released in public sources checked for this artifact.
- Debris: Blue Origin warned that debris from the hotfire anomaly may wash ashore and advised the public not to touch or approach it.

3. Frame Sequence Evidence

Frame-selection rule: The first visible brightening and immediately following pre-saturation expansion carry the highest geometry value. Once the scene saturates, later mushroom-cloud morphology is visually dramatic but weak for origin localization.

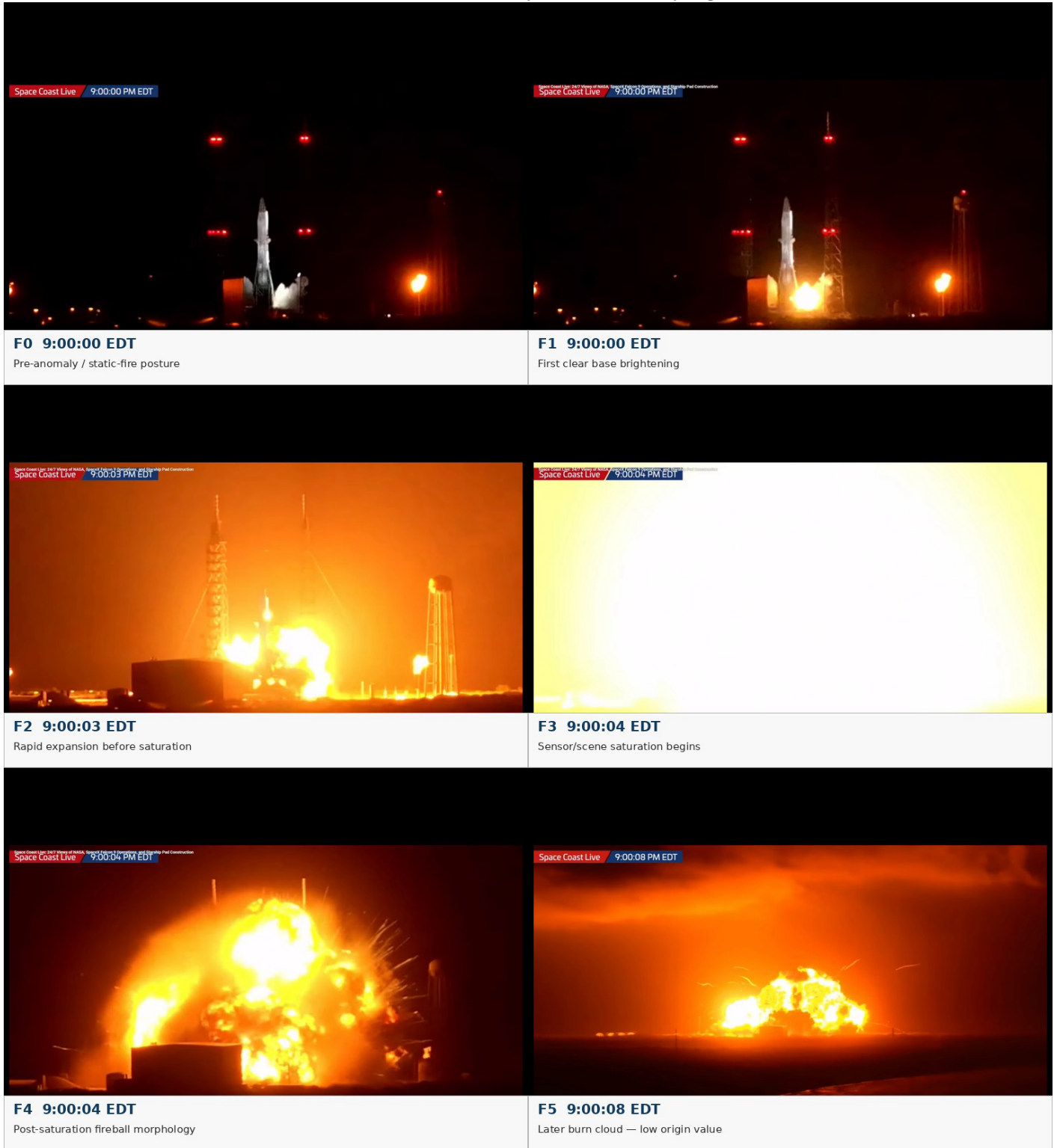


Figure 1. User-provided frame sequence, cropped to active video band. Later saturated frames are retained for sequence context but de-weighted for origin inference.

4. Annotated Camera-Plane Geometry

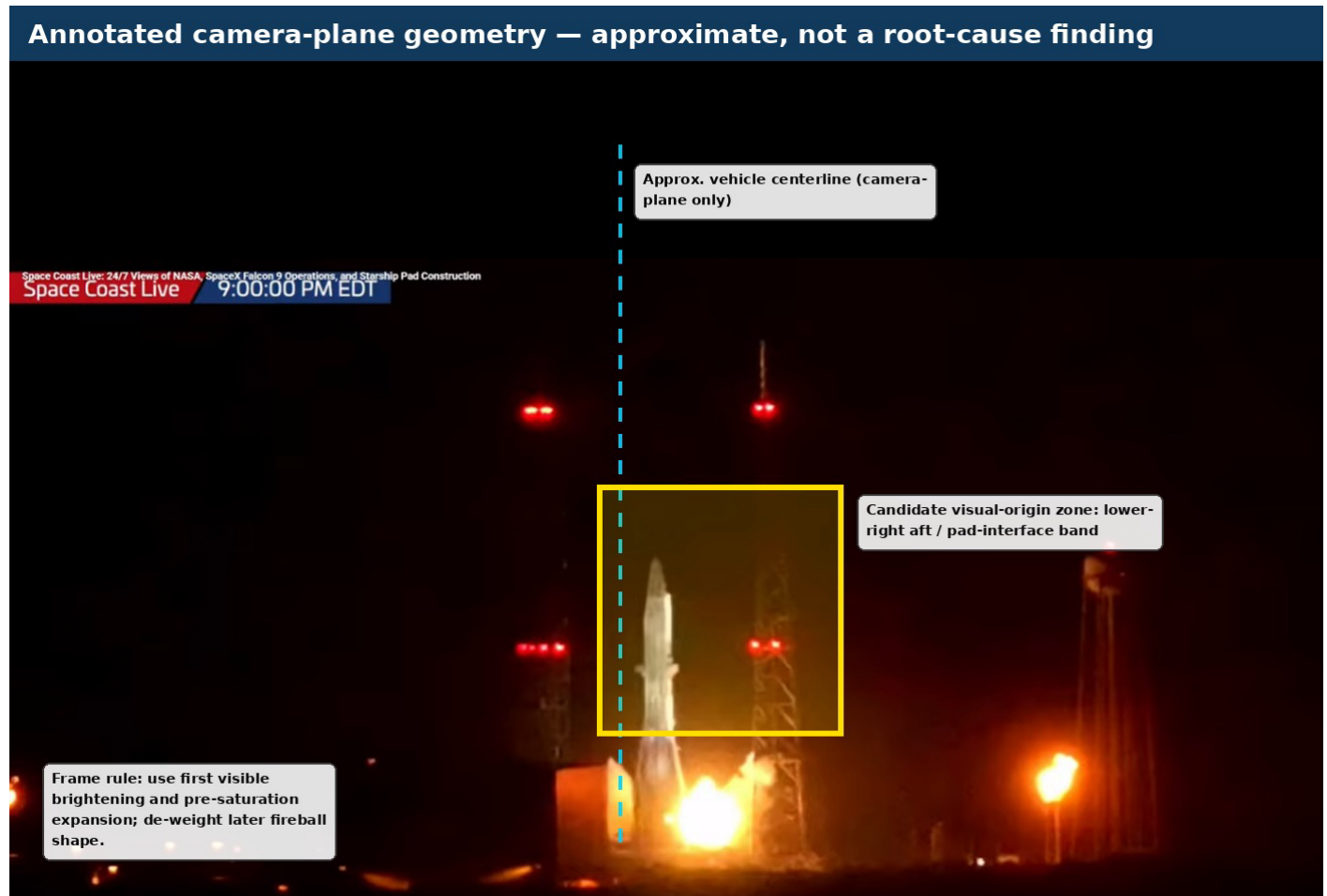


Figure 2. Approximate camera-plane annotation. The yellow box marks the candidate visual-origin band; it is not a finding of component, valve, seal, QD, engine, or GSE causation.

5. Revised Origin-Class Confidence Split

Important: These are visual-geometry priors only. They are not root-cause probabilities.

Class	Meaning	Revised geometry confidence	Reason
A	Inside projected vehicle envelope	20%	Some light overlaps the engine/aft vehicle projection, but image compression, distance, and exposure make true envelope placement uncertain.
B	On vehicle boundary / aft interface	40%	Most defensible visual class: lower-aft region adjacent to the projected vehicle boundary.
C	Outside vehicle boundary / pad-side interface	40%	Equally plausible from public frames because the first visible brightening sits near GSE / umbilical / trench-adjacent geometry in camera view.

6. Relative Mapping — What Can and Cannot Be Said

Reference	Assessment	Certainty
Vehicle centerline	First clear brightening appears slightly to camera-right of centerline at the lower aft region.	Moderate
Transporter/erector / tower geometry	Visible but dark and partly obscured; cannot reliably map first release to specific tower-side hardware.	Low
Umbilical / QD region	Plausible proximity from camera-plane geometry; not confirmable without pad-coordinate map and telemetry.	Low
Engine skirt / aft bay	Plausible because the visual anomaly is low enough to overlap aft-propulsion projection.	Low-to-moderate
Flame trench / deflector	Cannot be excluded. Initial appearance does not look purely trench-centered, but the camera view is insufficient.	Low
Hold-down / launch mount points	Possible adjacency; individual hardware is not resolvable in the uploaded frames.	Low

7. Evidence Needed Before Any Root-Cause Claim

- Vehicle and GSE telemetry synchronized to frame time: engine-start sequence, valve positions, abort commands, pressure/temperature excursions, mass-flow data, and leak-detector logs.
- Pad-coordinate geometry: camera location, lens metadata, vehicle/pad azimuth, hold-down / QD / umbilical positions, and lighting source map.
- Damage map: launch mount, tower, GSE piping, trench/deflector, engine section, tanks, umbilicals, and debris field.
- Multi-camera/video evidence: synchronized footage from independent angles before sensor saturation.
- Official investigation statements: Blue Origin, FAA, launch site authorities, and any regulatory findings.

8. Recommended Public-Facing Summary

Blue Origin's New Glenn suffered a static-fire / hotfire anomaly at LC-36 on May 28, 2026. From the public frame sequence, the earliest visible luminous anomaly appears low in the aft area, slightly to camera-right of the vehicle centerline, near the vehicle/pad interface band. That is a camera-plane geometry observation only. It does not establish whether the initiating failure was vehicle-side, pad/GSE-side, engine-bay, umbilical/QD, or trench-related. Root cause remains unresolved pending official investigation and telemetry.

9. Sources Checked / Source Discipline

User-provided frame sequence and v1 artifact: Primary visual material for this audit; includes social repost screenshots and Space Coast Live / NSF lower-third frames.

Associated Press, May 29, 2026: Public reporting on test firing, pad damage assessment, debris warning, and absence of payload aboard.

Business Insider, May 29, 2026: Public reporting on timing, Blue Origin statement, no injuries/fatalities, debris caution, and no payload aboard.

Blue Origin official New Glenn page: General New Glenn vehicle description and payload capability context.

Blue Origin integrated vehicle hotfire release: Context that integrated hotfire testing involves vehicle and ground systems in fully integrated on-pad configuration; not used for root-cause assignment.

Blue Origin second-stage hotfire release: Vehicle dimension and engine context: New Glenn above 320 ft; first stage powered by seven BE-4 engines.

10. Required Disclaimer

Disclaimer: Rocket Analyzer v6 is an AI-based simulation and diagnostic assistant. All outputs are advisory only. Certainty scores reflect inference confidence from public imagery and public reporting, not empirical proof. This artifact does not access real-time telemetry, proprietary data, internal investigation records, or regulatory findings and must not be used as a substitute for engineering review, regulatory compliance, or internal investigation procedures.

Grounded DI LLC

Artifact seal: $\Delta H = 0.0041$ | Tier-16 audit posture | Root-cause unresolved.